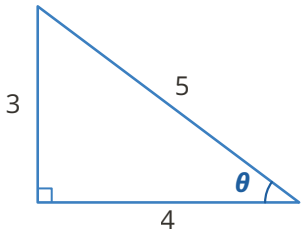
## The Fractal Five · Year 10 · new question types

Every new family (and the new variants of the reused ones), drawn at the printed size. Each cell shows when it is used, the formula strip as printed, and the answer computed from the same numbers that label the figure. Diagrams are drawn to scale.

### A RIGHT-ANGLED TRIANGLES AND SIMILAR FIGURES

trig_ratio · Terms 1–2 (formula printed) · (3, 4, 5, 'tan', 'h', 0)

**Q** Write tan $\theta$ as a fraction.

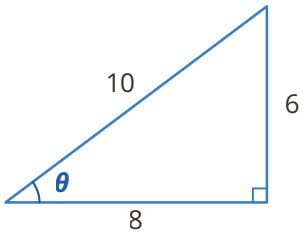



FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

ANSWER 3/4

trig_ratio · Terms 3–4 (from memory) · (6, 8, 10, 'sin', 'h', 1)

**Q** Write sin $\theta$ as a fraction.

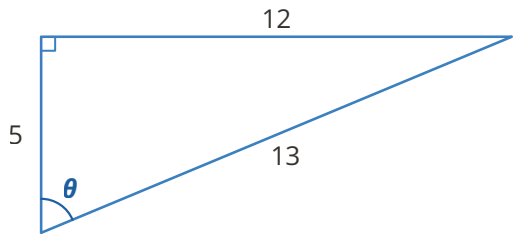



Formula recalled from memory

ANSWER 3/5

trig_ratio · Terms 1–2 (formula printed) · (12, 5, 13, 'cos', 'v', 2)

**Q** Write cos $\theta$ as a fraction.

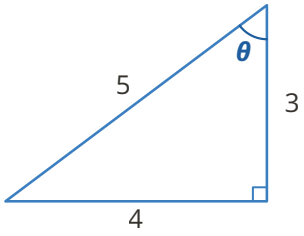



FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

ANSWER 5/13

trig_ratio · Terms 1–2 (formula printed) · (4, 3, 5, 'sin', 'v', 1)

**Q** Write sin $\theta$ as a fraction.

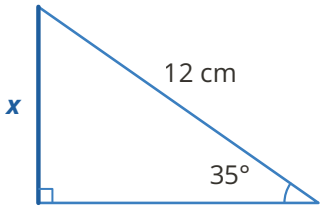



FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

ANSWER 4/5

trig_top · Terms 1–2 (formula printed) · ('sin', 12, 35, 'cm', 'h', 0)

**Q** Find $x$, correct to 1 decimal place.

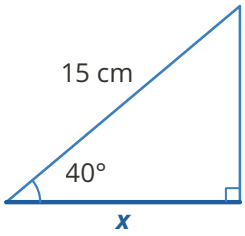



FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

ANSWER 6.9 cm

trig_top · Terms 1–2 (formula printed) · ('cos', 15, 40, 'cm', 'h', 1)

**Q** Find $x$, correct to 1 decimal place.

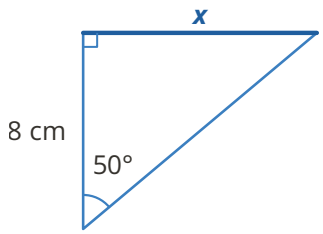



FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

ANSWER 11.5 cm

trig_top · Terms 1–2 (formula printed) · ('tan', 8, 50, 'cm', 'v', 2)

**Q** Find $x$, correct to 1 decimal place.

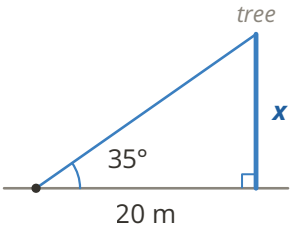



FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

ANSWER 9.5 cm

trig_top · Terms 3–4 (from memory) · ('elev', 35, 20, 'm', 'tree', 'R')

**Q** The angle of elevation is 35°. Find $x$, correct to 1 decimal place.

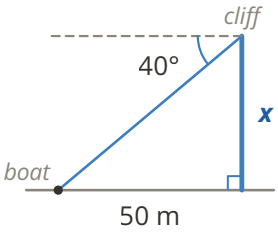



Formula recalled from memory

ANSWER 14.0 m

trig_top · Terms 3–4 (from memory) · ('dep', 40, 50, 'm', 'cliff', 'R')

**Q** The angle of depression is 40°. Find $x$, correct to 1 decimal place.

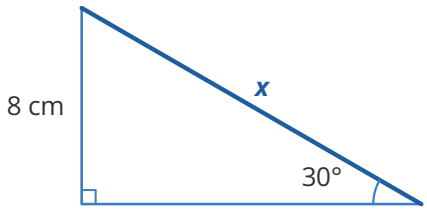



Formula recalled from memory

ANSWER 42.0 m

trig_bottom · Terms 1–2 (formula printed) · ('sin', 8, 30, 'cm', 'h', 0)

**Q** Find $x$, correct to 1 decimal place.

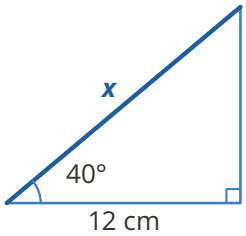



FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

ANSWER 16.0 cm

trig_bottom · Terms 1–2 (formula printed) · ('cos', 12, 40, 'cm', 'h', 1)

**Q** Find $x$, correct to 1 decimal place.

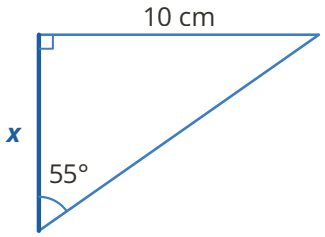



FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

ANSWER 15.7 cm

trig_bottom · Terms 1–2 (formula printed) · ('tan', 10, 55, 'cm', 'v', 2)

**Q** Find $x$, correct to 1 decimal place.

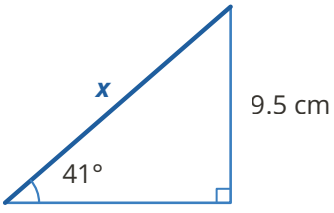



FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

ANSWER 7.0 cm

trig_bottom · Terms 3–4 (from memory) · ('sin', 9.5, 41, 'cm', 'h', 1)

**Q** Find $x$, correct to 1 decimal place.

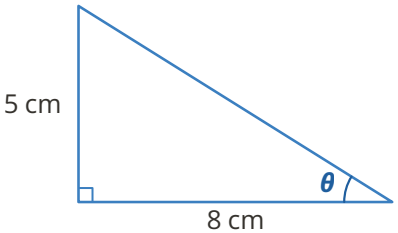



Formula recalled from memory

ANSWER 14.5 cm

trig_angle · Terms 1–2 (formula printed) · ('tan', 5, 8, 'cm', 'h', 0)

**Q** Find $\theta$, correct to the nearest degree.

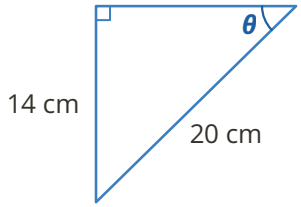



FORMULA $\theta = \sin^{-1}\frac{O}{H}$, $\cos^{-1}\frac{A}{H}$ or $\tan^{-1}\frac{O}{A}$

ANSWER 32°

trig_angle · Terms 3–4 (from memory) · ('sin', 14, 20, 'cm', 'h', 2)

**Q** Find $\theta$, correct to the nearest degree.

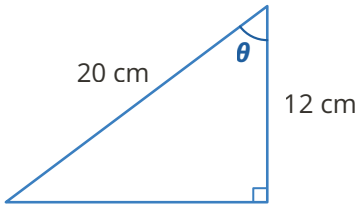



Formula recalled from memory

ANSWER 44°

trig_angle · Terms 3–4 (from memory) · ('cos', 12, 20, 'cm', 'v', 1)

**Q** Find $\theta$, correct to the nearest degree.

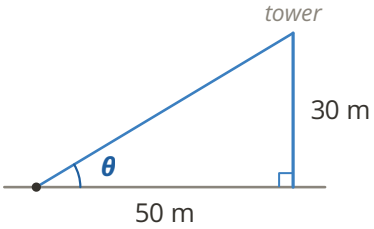



Formula recalled from memory

ANSWER 53°

trig_angle · Terms 3–4 (from memory) · ('elev', 30, 50, 'm', 'tower', 'R')

**Q** Find the angle of elevation, $\theta$, correct to the nearest degree.

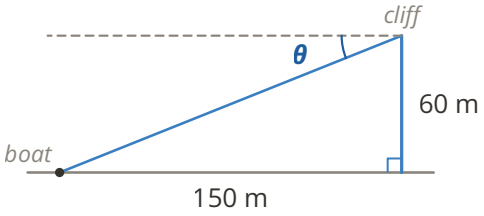



Formula recalled from memory

ANSWER 31°

trig_angle · Terms 3–4 (from memory) · ('dep', 60, 150, 'm', 'cliff', 'R')

**Q** Find the angle of depression, $\theta$, correct to the nearest degree.

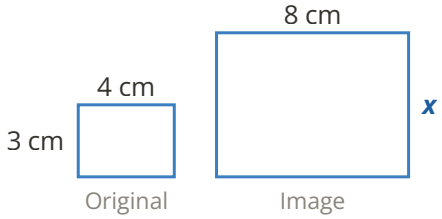



Formula recalled from memory

ANSWER 22°

similar · Terms 1–2 (formula printed) · (4, 3, 2, 'img', 'cm')

**Q** These rectangles are similar. Find the value of $x$.

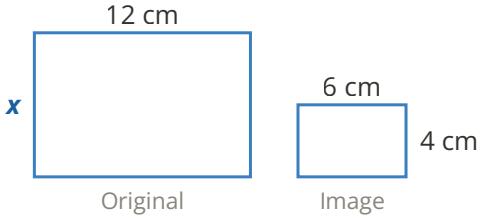



FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$

ANSWER 6 cm

similar · Terms 3–4 (from memory) · (12, 8, 0.5, 'org', 'cm')

**Q** These rectangles are similar. Find the value of $x$.

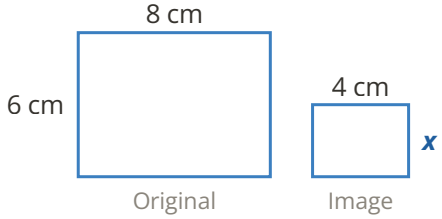



Formula recalled from memory

ANSWER 8 cm

similar · Terms 3–4 (from memory) · (8, 6, 0.5, 'img', 'cm')

**Q** These rectangles are similar. Find the value of $x$.

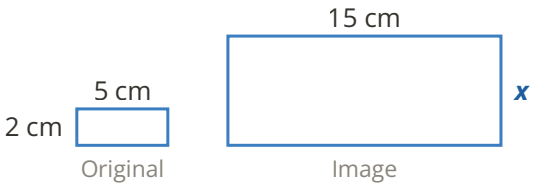



Formula recalled from memory

ANSWER 3 cm

similar · Terms 1–2 (formula printed) · (5, 2, 3, 'img', 'cm')

**Q** These rectangles are similar. Find the value of $x$.

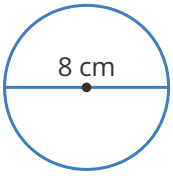



FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$

ANSWER 6 cm

a_circle · Terms 1–2 (formula printed) · ('d', 8, 'cm')

**Q** Find the area of this circle, correct to 1 decimal place.

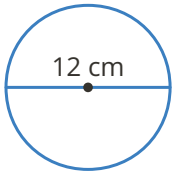



FORMULA $A = \pi r^2$

ANSWER 50.3 cm$^2$

a_circle · Terms 3–4 (from memory) · ('d', 12, 'cm')

**Q** Find the area of this circle, correct to 1 decimal place.

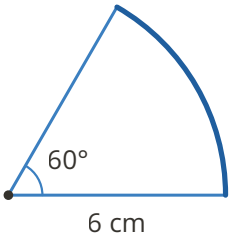



Formula recalled from memory

ANSWER 113.1 cm$^2$

arc · Terms 1–2 (formula printed) · (60, 6, 'cm')

**Q** Find the arc length, correct to 1 decimal place.

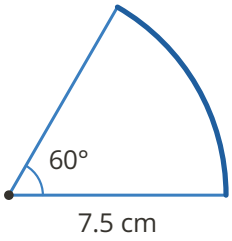



FORMULA $l = \frac{\theta}{360} \times 2\pi r$

ANSWER 6.3 cm

arc · Terms 3–4 (from memory) · (60, 7.5, 'cm')

**Q** Find the arc length, correct to 1 decimal place.

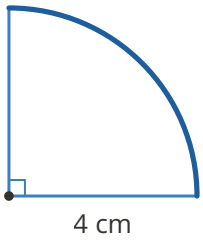



Formula recalled from memory

ANSWER 7.9 cm

arc · Terms 1–2 (formula printed) · (90, 4, 'cm')

**Q** Find the arc length, correct to 1 decimal place.

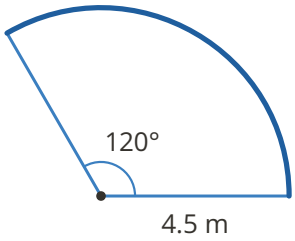



FORMULA $l = \frac{\theta}{360} \times 2\pi r$

ANSWER 6.3 cm

arc · Terms 3–4 (from memory) · (120, 4.5, 'm')

**Q** Find the arc length, correct to 1 decimal place.

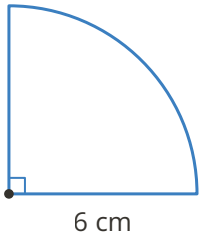



Formula recalled from memory

ANSWER 9.4 m

sector · Terms 1–2 (formula printed) · (90, 6, 'cm')

**Q** Find the area of this sector, correct to 1 decimal place.

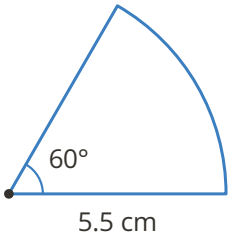



FORMULA $A = \frac{\theta}{360} \times \pi r^2$

ANSWER 28.3 cm$^2$

sector · Terms 3–4 (from memory) · (60, 5.5, 'cm')

**Q** Find the area of this sector, correct to 1 decimal place.

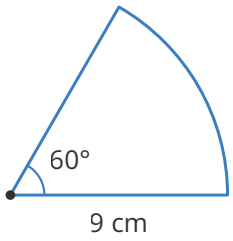



Formula recalled from memory

ANSWER 15.8 cm$^2$

sector · Terms 1–2 (formula printed) · (60, 9, 'cm')

**Q** Find the area of this sector, correct to 1 decimal place.

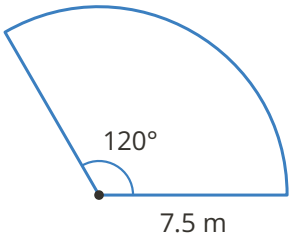



FORMULA $A = \frac{\theta}{360} \times \pi r^2$

ANSWER 42.4 cm²

sector · Terms 3–4 (from memory) · (120, 7.5, 'm')

**Q** Find the area of this sector, correct to 1 decimal place.

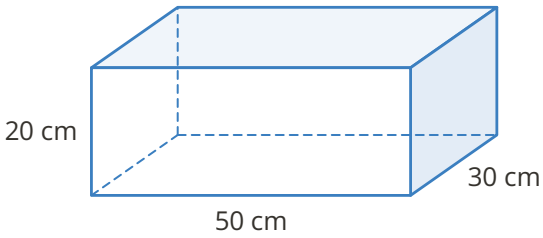



Formula recalled from memory

ANSWER 58.9 m²

capacity · Terms 1–2 (formula printed) · (50, 30, 20)

**Q** How many litres does this tank hold?

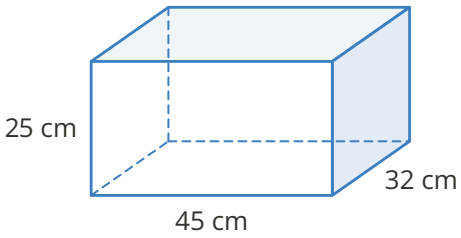



FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$

ANSWER 30 L

capacity · Terms 3–4 (from memory) · (45, 32, 25)

**Q** How many litres does this tank hold?

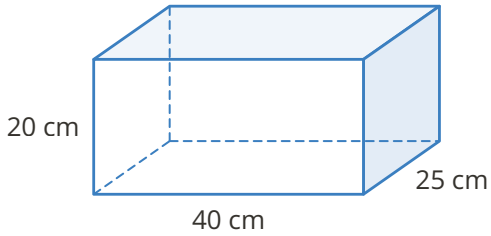



Formula recalled from memory

ANSWER 36 L

capacity · Terms 1–2 (formula printed) · (40, 25, 20)

**Q** How many litres does this tank hold?

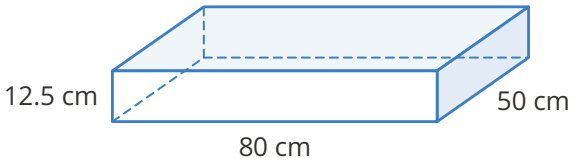



FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$

ANSWER 20 L

capacity · Terms 3–4 (from memory) · (80, 50, 12.5)

**Q** How many litres does this tank hold?

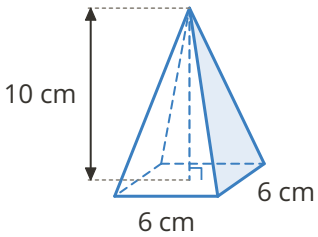



Formula recalled from memory

ANSWER 50 L

v_pyramid · Term 4 only, formula printed · (6, 6, 10, 'cm')

**Q** Find the volume of this pyramid.

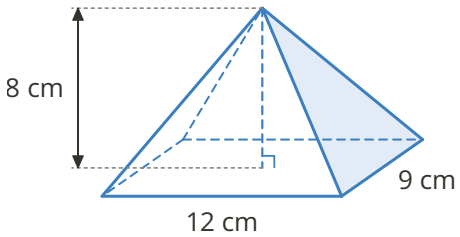



FORMULA $V = \frac{1}{3}Ah$ ($A$ = area of the base)

ANSWER 120 cm³

v_pyramid · Term 4 only, formula printed · (12, 9, 8, 'cm')

**Q** Find the volume of this pyramid.

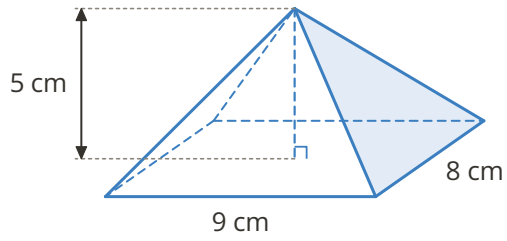



FORMULA $V = \frac{1}{3}Ah$ ($A$ = area of the base)

ANSWER 288 cm³

v_pyramid · Term 4 only, formula printed · (9, 8, 5, 'cm')

**Q** Find the volume of this pyramid.

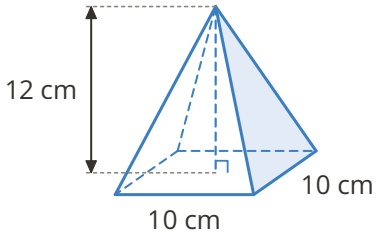



FORMULA $V = \frac{1}{3}Ah$ ($A$ = area of the base)

ANSWER 120 cm³

v_pyramid · Term 4 only, formula printed · (10, 10, 12, 'cm')

**Q** Find the volume of this pyramid.

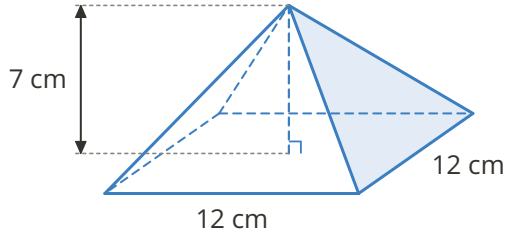



FORMULA $V = \frac{1}{3}Ah$ ($A$ = area of the base)

ANSWER 400 cm³

v_pyramid · Term 4 only, formula printed · (12, 12, 7, 'cm')

**Q** Find the volume of this pyramid.

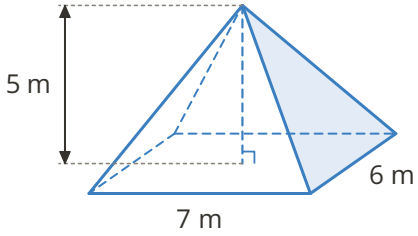



FORMULA $V = \frac{1}{3}Ah$ ($A$ = area of the base)

ANSWER 336 cm³

v_pyramid · Term 4 only, formula printed · (7, 6, 5, 'm')

**Q** Find the volume of this pyramid.

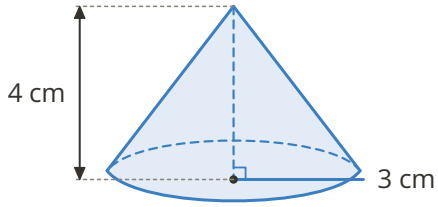



FORMULA $V = \frac{1}{3}Ah$ ($A$ = area of the base)

ANSWER 70 m³

v_cone · Term 4 only, formula printed · (3, 4, False, 'cm')

**Q** Find the volume of this cone, correct to 1 decimal place.

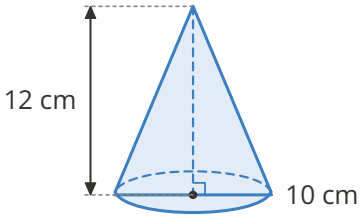



FORMULA $V = \frac{1}{3}\pi r^2 h$

ANSWER 37.7 cm³

v_cone · Term 4 only, formula printed · (5, 12, True, 'cm')

**Q** Find the volume of this cone, correct to 1 decimal place.

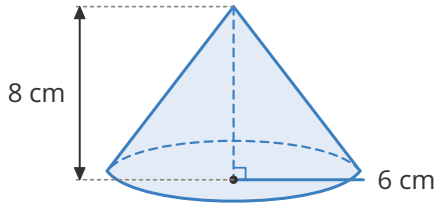



FORMULA $V = \frac{1}{3}\pi r^2 h$

ANSWER 314.2 cm³

v_cone · Term 4 only, formula printed · (6, 8, False, 'cm')

**Q** Find the volume of this cone, correct to 1 decimal place.

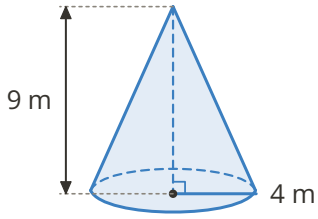



FORMULA $V = \frac{1}{3}\pi r^2 h$

ANSWER 301.6 $cm^3$

v_cone · Term 4 only, formula printed · (4, 9, False, 'm')

**Q** Find the volume of this cone, correct to 1 decimal place.

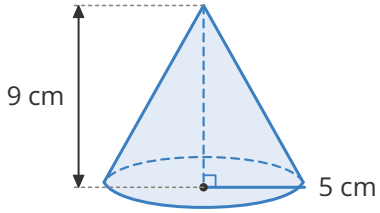



FORMULA $V = \frac{1}{3}\pi r^2 h$

ANSWER 150.8 $m^3$

v_cone · Term 4 only, formula printed · (5, 9, False, 'cm')

**Q** Find the volume of this cone, correct to 1 decimal place.

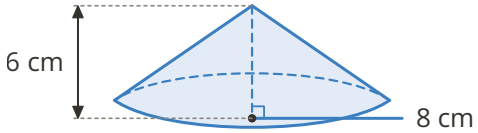



FORMULA $V = \frac{1}{3}\pi r^2 h$

ANSWER 235.6 $cm^3$

v_cone · Term 4 only, formula printed · (8, 6, False, 'cm')

**Q** Find the volume of this cone, correct to 1 decimal place.




FORMULA $V = \frac{1}{3}\pi r^2 h$

ANSWER 402.1 $cm^3$

## C PERCENTAGES AND MONEY

profit_pct · Terms 1–2 (formula printed) · (40, 50)

**Q** Find the percentage profit.

| | |
|---|---|
| Cost price | $40 |
| Selling price | $50 |

FORMULA Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

ANSWER 25%

profit_pct · Terms 1–2 (formula printed) · (80, 60)

**Q** Find the percentage loss.

| | |
|---|---|
| Cost price | $80 |
| Selling price | $60 |

FORMULA Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$

ANSWER 25%

profit_pct · Terms 3–4 (from memory) · (64, 80)

**Q** Find the percentage profit.

| | |
|---|---|
| Cost price | $64 |
| Selling price | $80 |

Formula recalled from memory

ANSWER 25%

profit_pct · Terms 3–4 (from memory) · (85, 68)

**Q** Find the percentage loss.

| | |
|---|---|
| Cost price | $85 |
| Selling price | $68 |

Formula recalled from memory

ANSWER 20%

interest · Terms 1–2 (formula printed) · (2000, 4, 3, 'years')

**Q** Find the simple interest on $2000 invested at 4% per year for 3 years.

**FORMULA** $I = Prn$ (interest = amount × yearly interest rate × number of years)

**ANSWER** $240

interest · Terms 3–4 (from memory) · (2000, 4, 2.5, 'years')

**Q** Find the simple interest on $2000 invested at 4% per year for 2.5 years.

Formula recalled from memory

**ANSWER** $200

interest · Terms 1–2 (formula printed) · (500, 5, 2, 'years')

**Q** Find the simple interest on $500 invested at 5% per year for 2 years.

**FORMULA** $I = Prn$ (interest = amount × yearly interest rate × number of years)

**ANSWER** $50

interest · Terms 3–4 (from memory) · (5000, 3, 18, 'months')

**Q** Find the simple interest on $5000 invested at 3% per year for 18 months.

Formula recalled from memory

**ANSWER** $225

wages · Terms 1–2 (formula printed) · ('Ava', 25, 30, 0, None)

**Q** Ava earns $25 an hour and works 30 hours. How much does Ava earn?

**FORMULA** Pay = hourly rate × hours

**ANSWER** $750

wages · Terms 1–2 (formula printed) · ('Liam', 24, 38, 4, 1.5)

**Q** Find Liam's total pay for the week.

| | |
|---|---|
| Hourly rate | $24 |
| Normal hours | 38 |
| Overtime hours (time-and-a-half) | 4 |

**FORMULA** Pay = hourly rate × hours; time-and-a-half = 1.5 × hourly rate

**ANSWER** $1056

wages · Terms 1–2 (formula printed) · ('Mia', 30, 35, 3, 2)

**Q** Find Mia's total pay for the week.

| | |
|---|---|
| Hourly rate | $30 |
| Normal hours | 35 |
| Overtime hours (double time) | 3 |

**FORMULA** Pay = hourly rate × hours; double time = 2 × hourly rate

**ANSWER** $1230

wages · Terms 3–4 (from memory) · ('Ava', 27.5, 30, 0, None)

**Q** Ava earns $27.50 an hour and works 30 hours. How much does Ava earn?

Formula recalled from memory

**ANSWER** $825

salary · Terms 1–2 (formula printed) · ('Sam', 52000, 'w')

**Q** Sam earns a salary of $52 000 a year. Find Sam's weekly pay.

FORMULA Weekly pay = yearly salary ÷ 52

ANSWER $1000

salary · Terms 3–4 (from memory) · ('Priya', 74250, 'f')

**Q** Priya earns a salary of $74 250 a year. Find Priya's fortnightly pay, to the nearest cent.

Formula recalled from memory

ANSWER $2855.77

salary · Terms 3–4 (from memory) · ('Sam', 68500, 'w')

**Q** Sam earns a salary of $68 500 a year. Find Sam's weekly pay, to the nearest cent.

Formula recalled from memory

ANSWER $1317.31

salary · Terms 1–2 (formula printed) · ('Priya', 62400, 'w')

**Q** Priya earns a salary of $62 400 a year. Find Priya's weekly pay.

FORMULA Weekly pay = yearly salary ÷ 52

ANSWER $1200

commission · Terms 1–2 (formula printed) · ('Mia', 5, 8000)

**Q** Find Mia's commission.

| | |
|---|---|
| Sales | $8000 |
| Commission rate | 5% |

FORMULA Commission = percentage × value of sales

ANSWER $400

commission · Terms 3–4 (from memory) · ('base', 'Mia', 400, 3, 12000)

**Q** Find Mia's total pay for the week.

| | |
|---|---|
| Weekly base wage | $400 |
| Commission | 3% of sales |
| Sales this week | $12 000 |

Formula recalled from memory

ANSWER $760

commission · Terms 1–2 (formula printed) · ('Josh', 3, 12000)

**Q** Find Josh's commission.

| | |
|---|---|
| Sales | $12 000 |
| Commission rate | 3% |

FORMULA Commission = percentage × value of sales

ANSWER $360

commission · Terms 3–4 (from memory) · ('base', 'Josh', 350, 5, 6400)

**Q** Find Josh's total pay for the week.

| | |
|---|---|
| Weekly base wage | $350 |
| Commission | 5% of sales |
| Sales this week | $6400 |

Formula recalled from memory

ANSWER $670

holiday_pay · Terms 1–2 (formula printed) · ('Jo', 1200)

**Q** Jo earns $1200 a week. Find Jo's total holiday pay for 4 weeks, including 17.5% leave loading.

**FORMULA** Holiday pay = 4 weeks' pay + 17.5% of 4 weeks' pay

**ANSWER** $5640

holiday_pay · Terms 3–4 (from memory) · ('Jo', 1135)

**Q** Jo earns $1135 a week. Find Jo's total holiday pay for 4 weeks, including 17.5% leave loading.

Formula recalled from memory

**ANSWER** $5334.50

holiday_pay · Terms 1–2 (formula printed) · ('Ari', 800)

**Q** Ari earns $800 a week. Find Ari's total holiday pay for 4 weeks, including 17.5% leave loading.

**FORMULA** Holiday pay = 4 weeks' pay + 17.5% of 4 weeks' pay

**ANSWER** $3760

holiday_pay · Terms 3–4 (from memory) · ('Ari', 1286.4)

**Q** Ari earns $1286.40 a week. Find Ari's total holiday pay for 4 weeks, including 17.5% leave loading.

Formula recalled from memory

**ANSWER** $6046.08

## D INDICES, FORMULAS AND RATES

idx · Terms 1–2 (formula printed) · ('mult', 'x', 4, 3, 3, 2)

**Q** Simplify.

$$3x^4 \times 2x^3$$

**FORMULA** $a^m \times a^n = a^{m+n}$

**ANSWER** $6x^7$

idx · Terms 1–2 (formula printed) · ('div', 'a', 5, 2, 12, 4)

**Q** Simplify.

$$12a^5 \div 4a^2$$

**FORMULA** $a^m \div a^n = a^{m-n}$

**ANSWER** $3a^3$

idx · Terms 3–4 (from memory) · ('pow', 'a', 3, 2, 2, 1)

**Q** Simplify.

$$(2a^3)^2$$

Formula recalled from memory

**ANSWER** $4a^6$

zero_neg · Terms 1–2 (formula printed) · ('zero', 'plus', 7, 3)

**Q** Evaluate.

$$7^0 + 3$$

**FORMULA** $a^0 = 1$

**ANSWER** 4

zero_neg · Terms 3–4 (from memory) · ('neg', 2, 5)

Q Evaluate. Write your answer as a fraction.

$2^{-5}$

Formula recalled from memory

ANSWER 1/32

zero_neg · Terms 1–2 (formula printed) · ('zero', 'pow', 5)

Q Evaluate.

$5^{0}$

FORMULA $a^0 = 1$

ANSWER 1

zero_neg · Terms 3–4 (from memory) · ('neg', 6, 2)

Q Evaluate. Write your answer as a fraction.

$6^{-2}$

Formula recalled from memory

ANSWER 1/36

sci · Terms 1–2 (formula printed) · ('to', 45000)

Q Write in scientific notation.

45 000

FORMULA $a \times 10^n$, where $1 \le a < 10$

ANSWER $4.5 \times 10^4$

sci · Terms 3–4 (from memory) · ('from', 4.5, -2)

Q Write as a basic numeral.

$4.5 \times 10^{-2}$

Formula recalled from memory

ANSWER 0.045

sci · Terms 3–4 (from memory) · ('to', 0.0062)

Q Write in scientific notation.

0.0062

Formula recalled from memory

ANSWER $6.2 \times 10^{-3}$

sci · Terms 1–2 (formula printed) · ('to', 3200000)

Q Write in scientific notation.

3 200 000

FORMULA $a \times 10^n$, where $1 \le a < 10$

ANSWER $3.2 \times 10^6$

formula_sub · Terms 1–2 (formula printed) · ('vuat', 'v', {'u': 4, 'a': 3, 't&#

Q Find $v$ when $u = 4$, $a = 3$ and $t = 5$.

$v = u + at$

FORMULA $v = u + at$

ANSWER 19

formula_sub · Terms 1–2 (formula printed) · ('perim', 'P', {'l': 8, 'w': 5})

Q Find $P$ when $l = 8$ and $w = 5$.

$$P = 2(l + w)$$

FORMULA $P = 2(l + w)$

ANSWER 26

formula_sub · Terms 3–4 (from memory) · ('cost', 'd', {'C': 17})

Q Find $d$ when $C = 17$.

$$C = 5 + 2.4d$$

Formula recalled from memory

ANSWER 5

formula_sub · Terms 3–4 (from memory) · ('temp', 'C', {'F': 68})

Q Find $C$ when $F = 68$.

$$F = 1.8C + 32$$

Formula recalled from memory

ANSWER 20

formula_sub · Terms 3–4 (from memory) · ('tri', 'h', {'A': 18, 'b': 4.5})

Q Find $h$ when $A = 18$ and $b = 4.5$.

$$A = ½bh$$

Formula recalled from memory

ANSWER 8

formula_sub · Terms 3–4 (from memory) · ('line', 'x', {'y': -7, 'm': 2, 'c&

Q Find $x$ when $y = −7$, $m = 2$ and $c = 3$.

$$y = mx + c$$

Formula recalled from memory

ANSWER −5

ratio · Terms 1–2 (formula printed) · (120, 3, 5, '$')

Q Share $120 in the ratio 3 : 5.

FORMULA One part = amount ÷ total number of parts

ANSWER $45 and $75

ratio · Terms 3–4 (from memory) · (75, 2, 3, '$')

**Q** Share $75 in the ratio 2 : 3.

Formula recalled from memory

**ANSWER** $30 and $45

ratio · Terms 1–2 (formula printed) · (200, 3, 2, '$')

**Q** Share $200 in the ratio 3 : 2.

**FORMULA** One part = amount ÷ total number of parts

**ANSWER** $120 and $80

ratio · Terms 3–4 (from memory) · (1250, 3, 7, '$')

**Q** Share $1250 in the ratio 3 : 7.

Formula recalled from memory

**ANSWER** $375 and $875

## E STRAIGHT-LINE GRAPHS, DATA AND PROBABILITY

linear · Terms 1–2 (formula printed) · ('si', 3, -2)

**Q** State the gradient and the *y*-intercept.

$$y = 3x - 2$$

**FORMULA** $y = mx + c$ ($m$ = gradient, $c$ = $y$-intercept)

**ANSWER** gradient 3, *y*-intercept −2

linear · Terms 3–4 (from memory) · ('si', -2, 7)

**Q** State the gradient and the *y*-intercept.

$$y = -2x + 7$$

Formula recalled from memory

**ANSWER** gradient −2, *y*-intercept 7

mean · Terms 1–2 (formula printed) · ('freq', (2, 3, 4, 5, 6), (1, 3, 2, 3, 1))

**Q** Find the mean score.

| Score | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| Frequency | 1 | 3 | 2 | 3 | 1 |

**FORMULA** Mean = $\frac{\Sigma fx}{\Sigma f}$

**ANSWER** 4

mean · Terms 3–4 (from memory) · ('freq', (1, 2, 3, 4, 5), (3, 5, 8, 2, 1))

**Q** Find the mean score, correct to 1 decimal place.

| Score | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Frequency | 3 | 5 | 8 | 2 | 1 |

Formula recalled from memory

**ANSWER** 2.6

freq · Terms 1–2 (formula printed) · ('exp', 'A fair die is rolled 300 times. How many 4s would y

**Q** A fair die is rolled 300 times. How many 4s would you expect?

FORMULA Expected frequency = $P(E) \times$ number of trials

ANSWER 50

freq · Terms 3–4 (from memory) · ('rel', 'A coin is tossed 80 times and lands heads 44 times. Fi

**Q** A coin is tossed 80 times and lands heads 44 times. Find the relative frequency of heads.

Formula recalled from memory

ANSWER 0.55 (or 11/20)

freq · Terms 3–4 (from memory) · ('exp', 'The probability that a seed grows is 0.85. If 400 seed

**Q** The probability that a seed grows is 0.85. If 400 seeds are planted, how many would you expect to grow?

Formula recalled from memory

ANSWER 340

freq · Terms 1–2 (formula printed) · ('exp', 'A fair coin is tossed 200 times. How many heads w

**Q** A fair coin is tossed 200 times. How many heads would you expect?

FORMULA Expected frequency = $P(E) \times$ number of trials

ANSWER 100